

Molecular Biology Laboratory

Attending the labs is obligatory. If you do not come to the labs you cannot complete the practical part of the course. Arrive on time. If there is a justifiable reason for the delay - please inform the TA in advance or as soon as possible.

- You must come equipped with a lab coat and safety glasses; you must wear closed shoes. Each pair of students must come with printed protocol and their lab notebook. All of you must obey the safety instructions - including the current coronavirus transmission mitigation rules.
- We will expel students who do not follow the safety rules. If we expel you, you were not attending the lab. Attending the lab is part of your mark.

Work

- There are only two units this year because of the shortened semester. Both are wet labs: 1 - what is the genetic material, 2 - structures of genomes.
- You work in pairs. A set of 4 or 5 pairs is a team. Each team has a Teaching Assistant (TA).
- Students in a team decide on the course of the experiment, plan what each pair will do, considering the target of the research. Your TAs will not tell you what to do! They will answer questions about how things work and show you how to work properly on the bench.
- *The Lab Notebook*: A lab notebook is the cornerstone of research; it is your research diary. The cover of the notebook must show your names and contact information. Everything concerning your work and your thinking about your work is written or indicated there. This includes, what your group decided to do, how it decided, and why. It includes what you did, your observations while you were doing it, and your conclusions. It includes any problems or questions that arose while you were doing it. If you think you did something wrong - write it down in the notebook. It includes your results, what you think about your results, what your conclusions are and what you want to do next. If you googled something in the evening or thought of something while you fall asleep at night - it includes that too. Everything you write includes the date of writing it. If you are writing about yesterday's experiment, you write today's date and then write that the topic is yesterday's work. Do not leave blank pages. At the end of each lab session your TA will sign your notebook.

Grading

50% notebook, 25% bench performance, 25% take-home exam. You will submit your lab notebooks to your TA at the end of the course and it will be graded according to your compliance with the outlined goals of keeping a good lab notebook. A take-home exam will be posted to Moodle at the end of the course. It will include questions related to material learned in the lab. Each pair will submit one copy of the exam according to the posted rules.

Sample lab notebook page.

DATED

2/1/04

Samples labelled

our LB
mediumDroog's LB
medium

$$1 = (A2, A3, \text{Zla}, \text{Mtwi}) + (n, x)$$

$$2 = +, - \quad (+) \text{ induced}, (-) \text{ uninduced}$$

$$3 = \phi, \text{ washed} \quad (\phi \Rightarrow \text{null} = \text{new tubes})$$

washed = pre-washed, washed tubes

DATE OF SAMPLE PREPARATION

NOTE FROM OTHER DATE
ADDED LATER

Preparing samples from 2/1/04 for gel.

- Add 200 μ l Cracking buffer

Incubate 37°C, 30'

Add 200 μ l 2x SB. freeze -20°C

4/1/04

gel pics
2/1/04
Add A3 n
fused

Chromosomal stock

final concentration is 12.5 μ g/mlstock $\times 1000$ 12.5 μ g/ml $\Rightarrow$ EtOHprepare 1.5 ml 12.5 $\times 1.5 = 12.5$

$$\begin{array}{r} 6.25 \\ 18.75 \text{ mg} \end{array}$$

5/1/04

Gel load order